

James Williams

📞 754-230-2284 ✉️ ja159241@ucf.edu 🔗 [linkedin.com/james-williamsiv](https://www.linkedin.com/james-williamsiv) 🐙 github.com/jamesw7863

Education

University of Central Florida

Expected: Spring 2028

B.S. in Computer Science, Minor in Math (GPA: 3.7 / 4.0)

Orlando, FL

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Computer Architecture
- **Organizations & Clubs:** ColorStack, NSBE, KnightHacks, Events & Outreach Coordinator @ F1 AI Racer Research

Experience

New York Life

May 2026 – Aug 2026

Software Engineer Intern

New York, NY

- Modernized a legacy mainframe commission system with **React**, **Spring Boot**, **PostgreSQL**, and **Spring Batch**, reducing rule update effort by **50%** through database-driven configuration for business users
- Designed a rules-based payout workflow to process **10K+** agent and product records, improving commission processing flexibility by **40%** and supporting downstream integrations with **SAP** and **FUELS**
- Built a **Databricks** ELT pipeline with **PySpark** and **Amazon S3**, cutting silver-to-gold transformation time by **35%** through agentic AI code generation and human-in-the-loop PR review workflows

Institute for Simulation and Training – UCF

Dec 2025 – Jan 2026

Software Engineer Intern

Orlando, FL

- Optimized a **Unity** based digital twin of a nuclear glove box system using **C#**, simulating airflow, pressure, and fan-filter dynamics to support operator training and improve system comprehension by **30%**
- Designed a modular simulation architecture with fault injection and objective-based evaluation, enabling leak scenarios, pass/fail logic, and automated performance assessment under realistic operating conditions
- Collaborated with UCF's Mixed Emerging Technology Integration Lab on a **U.S. Department of Energy** sponsored project, translating nuclear facility requirements into an interactive simulation used by **10+** stakeholders

Brown and Brown Insurance

May 2025 – Aug 2025

Software Engineer Intern

Daytona Beach, FL

- Architected and tested an enterprise-level global dashboard using **Python**, **React**, **Leaflet.js**, and **PostgreSQL** to optimize infrastructure visibility across more than **700** international corporate office locations
- Enhanced outage response time by **40%** through the implementation of real-time geospatial tracking features, live system health monitoring, and map-based alert visualization tools to help operators respond more efficiently
- Collaborated in **Agile** sprints and contributed to a scalable backend pipeline for real-time global power tracking integrating **PostgreSQL**, **REST APIs**, and distributed data ingestion workflows across multiple regions

Projects

FirstStep | Next.js, TypeScript, Tailwind, Node.js, Web Scraping, Resend, Vercel

Apr 2026

- Built a **Next.js** internship discovery platform delivering **daily/weekly** scraped postings with direct application links
- Automated internship collection using **Node.js** and web scraping, reducing manual posting discovery time by **80%**
- Integrated **Resend** email digests and deployed on **Vercel**, helping **100+** students discover internship opportunities

InvestEd | Next.js, TypeScript, Zustand, shadCN, Gemini 2.5 Flash, Vercel, Tailwind

Oct 2025

- Placed **3rd** out of 28 teams in the Knight Hacks 2025 OneEthos AI for Financial Empowerment Challenge
- Engineered an AI chatbot with **Gemini 2.5 Flash API** to generate personalized investing advice
- Built a Smart HYSA Finder with APY rankings and an Investment Dashboard with asset performance analytics

MOVE | Next.js, TypeScript, Tailwind, WebGPU/WebGL, MediaPipe, TensorFlow, Computer Vision

Sep 2025

- Built a real-time computer vision game using **MediaPipe Pose** to track movement at 60 FPS with <80ms latency
- Developed game logic in **Next.js** to handle collisions, scoring, and a pausing feature when players leave the frame
- Implemented persistent high-score storage and responsive UI, tested by **40+** users during ShellHacks 2025

Technical Skills

Languages: Python, C/C++, C#, Java, MySQL, PostgreSQL, CSS, HTML, JavaScript, TypeScript

Technologies: React, Next.js, Node.js, Flask, Spring Boot, Tailwind CSS, Databricks, PySpark, AWS, Azure, GitHub

Concepts: Version Control, APIs, Front-End Development, Back-End Development, Machine Learning, Computer Vision